Career Pilot: An AI-Powered Resume Screener and Job Recommendation

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I. Abstract – The exponential growth of online job applications has made manual resume screening slow, inconsistent, and prone to bias. Traditional Applicant Tracking Systems (ATS) rely heavily on rigid keyword matching, often overlooking qualified candidates whose resumes use varied terminology. Recent advancements in AI and Natural Language Processing (NLP) have enabled automated parsing of unstructured resumes, extracting key details such as skills, education, and work experience into structured formats for faster comparison with job requirements. Approaches employing vector space models and cosine similarity, as well as transformer-based embeddings like BERT, have demonstrated improved contextual matching over simple keyword searches, while providing recruiters with candidate rankings, market trend analytics, and actionable recommendations for applicants. Complementary research on job skill recommendation systems, such as SkillRec, leverages large datasets of job postings and embedding techniques (BERT, FastText) to identify the most relevant skills for a given role, helping candidates bridge skill gaps and align more closely with market demands. By integrating AI-driven resume analysis, semantic similarity scoring, and skill recommendation, these systems collectively aim to enhance recruitment efficiency, improve matching accuracy, and create a fairer, data-driven hiring process that benefits both employers and job seekers.

Keywords
Artificial Intelligence, Machine Learning, Natural Language Processing, Sentence Transformer, BERT, Cosine Similarity, Neural Network, Resume Screening, Job Recommendation, Skill Recommendation, Web Application.

II. Introduction

In this project, we introduce our AI Resume Screener and Job Recommender, a platform we built to tackle the core inefficiencies and biases in modern hiring. The flood of applications from online job portals has overwhelmed traditional methods; manual screening is simply too slow, and older Applicant Tracking Systems (ATS) often fail because their rigid keyword matching overlooks great candidates. Our approach is different. We've designed a single, unified system that combines deep resume analysis, intelligent job matching, and proactive skill recommendations, creating a complete solution that serves both recruiters and job seekers. Our goal is not just to make screening faster and more accurate for companies, but also to empower candidates with personalized guidance for their own career growth.
We specifically designed our system to solve a common problem where resume analysis, matching, and skill recommendations are handled as separate, disconnected tasks. To bring them together, we use a SentenceTransformer('all-MiniLM-L6-v2') model, which allows our system to understand the context and meaning within a resume by generating semantic embeddings. It then calculates a precise match score using cosine similarity. In addition, we analyze a candidate's skillset against current market demands using a predefined knowledge base of job roles and skills to suggest relevant courses By combining these technologies into one platform, we believe our project offers a powerful tool to make the entire hiring lifecycle more efficient, transparent, and fair for everyone involved.

III. Literature Survey

The integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) into recruitment systems has been a subject of active research, aiming to address inefficiencies in manual resume screening and the limitations of traditional Applicant Tracking Systems (ATS). Existing literature reveals a range of approaches, from automated parsing and similarity matching to advanced recommendation and skill identification systems.

K.L. Abhishek proposed an AI-powered Resume Analyzer that automates resume parsing, analysis, and recommendation. The system uses Pyresparser and the Natural Language Toolkit (NLTK) to extract structured data—such as education, experience, and skills—from unstructured resume documents. A Streamlit-based interface enables recruiters to upload and review resumes, visualize analytics, and export structured candidate data in CSV format. Notably, the system offers dual benefits: it provides recruiters with insights into market trends and popular roles, while also giving applicants targeted recommendations to enhance their resumes. The study highlights the potential of combining NLP-driven extraction with recruiter-facing analytics and applicant feedback loops.[1]

In a similar domain, Chirag Daryani developed an Automated Resume Screening System using NLP and vector space modeling. Their two-phase approach first extracts relevant qualifications from resumes using tokenization, stemming, lemmatization, and named entity recognition. The second phase ranks resumes based on cosine similarity between the extracted content and the job description, producing an ordered shortlist of candidates. This approach overcomes significant ATS drawbacks by providing a semantic matching, allowing better selection of synonyms, contextual phrasing of queries, and eliminating different resume formats, therefore enhancing the degree of fairness. [2].

Expanding beyond screening, Nikhat Akhtar introduced an AI-Driven Intelligent Resume Recommendation Engine that leverages transformer-based language models such as BERT to match candidate resumes with job postings at a semantic level. The

system not only recommends the most relevant job opportunities but also predicts future roles a candidate might qualify for based on their current skill set. Additionally, it incorporates a feedback mechanism, enabling applicants to refine their resumes and align them more closely with industry requirements. This approach bridges the gap between candidate evaluation and career development support, making the recruitment process more interactive and future-focused.[3]

Ong and Lim addressed a related but distinct challenge: identifying the key skills required for specific job titles. Their SkillRec system collects and processes large datasets of job postings, extracting job titles and skill mentions to create a structured skills database. Using embedding techniques such as BERT and FastText, SkillRec models job title representations and applies a feed-forward neural network to recommend relevant skills. Tested on 6,000 job listings and 589 unique skills, the system demonstrated promising accuracy and F1-scores. By enabling candidates to identify and acquire in-demand skills, SkillRec serves as a complementary tool to resume screening systems, supporting long-term employability. [4]

Collectively, these studies [1]– [4] demonstrate that AI-based recruitment solutions are evolving toward comprehensive ecosystems that combine automated parsing, semantic matching, job recommendation, and skill analytics. However, most existing solutions focus on one or two aspects of the recruitment workflow.

Reference	Problem Addressed	Methodology	Findings	Accuracy	Significance to Present Work
[1]	Inefficiency and bias in manual resume screening; need for automated analysis and dual benefits for applicants and recruiters.	AI-powered tool using Pyresparser and NLTK for resume parsing; MySQL database; Streamlit-based UI; recruiter analytics and applicant recommendations.	Automates data extraction, delivers personalized resume improvement tips, and provides market trend analytics to recruiters.	~85 %	Establishes a foundation for automated parsing and inspires the dual applicant-recruiter benefit model in our system.
[2]	Time-consuming manual shortlisting; need for automated candidate ranking.	Two-phase pipeline: (1) NLP preprocessing (NER with spaCy) for JSON extraction; (2) Content-based ranking via Vector Space Model, TF-IDF, and cosine similarity.	Efficiently filters and ranks resumes based on similarity scores; demonstrates limitations of purely keyword-based filtering.	~83 %	Provides a baseline for similarity-based ranking and motivates using semantic embeddings (BERT) to improve upon it.
[3]	Poor, inaccurate resume recommendation; one requires semantic matching and feedback on candidates.	Data collection: The web crawler; resume similarity and job description similarity: the BERT similarities; resumes and job description Similarity: the web crawler; match scorings: cosine similarity; feedback and career prediction.	Deep learning models (BERT) are much better semantic matching options and provide meaningful feedback to candidates.	~89 %	Directly influences our system's core methodology of using BERT with cosine similarity and inspires integrated feedback features.
[4]	Lack of data-driven tools for job skill recommendations; candidates are often unaware of evolving skill demands.	Large-scale job posting collection; embeddings (BERT & FastText) for job titles; feed-forward neural network to recommend skills.	Embedding-based approach is highly effective for data-driven skill mapping and career guidance.	~84 %	Informs the skill-gap analysis and targeted recommendation component of our project.

- V. Methodology
- Project Goal: To develop a comprehensive AI Resume Screener and Job Recommendation system to streamline the hiring process for both job seekers and recruiters.
- Core Technology: The system is built as a unified web application integrating advanced Natural Language Processing (NLP), semantic matching, and skill recommendation.
- Resume-to-Job Matching: Model: Leverages the sentence-transformers library with the 'all-MiniLM-L6-v2' model to generate contextual embeddings for both resumes and job descriptions. Process: Generates contextual embeddings for both resumes and job descriptions. Metric: Uses cosine similarity to calculate a precise percentage match between a candidate's resume and a job opening.
- Skill Gap Analysis & Recommendation: Logic: Employs a CSV-based knowledge base to identify required skills

for a job category. It compares these with the skills extracted from the resume to identify gaps and recommend courses from the knowledge base Function: Analyzes a candidate's existing skills against required job skills to identify gaps. Output: Recommends relevant courses and learning paths to help candidates upskill.

- User Experience: The entire system will be accessible through a user-friendly web interface, allowing for easy resume uploads and personalized job and course recommendations.
- Platform Integration: Plans include integrating the system with major job portals like JSearch API (via RapidAPI) to fetch real-time job listings and enable direct applications.
- Resume Validation: To improve robustness and user experience, the system includes a heuristic-based validation function. Before processing, this function performs a preliminary check on the uploaded file's content. It scores the document based on the presence of common resume keywords (e.g., 'experience', 'education', 'skills') and penalizes it for keywords typically found in academic papers (e.g., 'abstract', 'references', 'conclusion'). This step ensures that only valid resumes are passed to the analysis engine, preventing errors from irrelevant file uploads.

VI. Gap Analysis

The project aims to develop a sophisticated AI-driven resume screening and job recommendation system, building upon established research in NLP and machine learning for recruitment. The foundational papers identified—Abhishek et al.'s intelligent resume screening tool [1], Daryani et al.'s automated resume screening system [2], Akhtar et al.'s AI-driven resume recommendation engine [3], and Ong and Lim's SkillRec for job skill recommendation [4]—provide a robust theoretical and practical bedrock. However, our proposed innovations introduce crucial capabilities that address existing gaps, particularly in holistic candidate development, real-time integration, and actionable insights.

Foundational Contributions from Referenced Papers:

1 Intelligent Resume Screening and Analytics [1]: Abhishek et al.'s paper lays out a practical approach to automating resume analysis, extracting key information, and providing analytics for recruiters, along with personalized feedback for applicants. This forms a vital part of our system's initial resume processing and user feedback mechanisms.

2 Automated Resume Screening and Similarity [2]: Daryani et al.'s work focuses on automating resume shortlisting using NLP techniques and TF-IDF with cosine similarity for ranking. This provides a baseline understanding of similarity-based matching and highlights the need for more advanced semantic approaches that our project will employ.

3 AI-Driven Resume Recommendation Engine [3]: Akhtar et al.'s research directly addresses the core of our project by utilizing BERT and cosine similarity for accurate semantic matching between resumes and job descriptions. Their work on providing percentage matches and predicting future job prospects is a direct inspiration for our job recommendation and feedback features.

4 Job Skill Recommendation for Career Insights [4]: Ong and Lim's paper demonstrates the effectiveness of embedding techniques (like BERT and FastText) combined with neural networks for recommending relevant job skills based on job titles. This provides a direct blueprint for our project's course recommendation module, which aims to help users develop in-demand skills.

This Project Innovations and Gap Analysis:

Our project extends these foundational capabilities in three significant ways, addressing key limitations and expanding the utility of AI in recruitment for practical applications:

- Holistic Candidate Development & Course Recommendation:

Gap: While papers [1] and [3] offer some feedback or future job prospects, none of the core papers directly integrate a robust system for recommending specific courses or learning paths to bridge skill gaps. Existing solutions often focus solely on job matching without providing actionable steps for career growth.

Innovation: This is confronted head on in our project by the fact that a pre-established body of knowledge is built into the project in the form of a CSV file. This file is used to match job categories with the required skills and the links with online courses. The system conducts a gap analysis, which involves comparing the skills held by the resume to this body of knowledge and consequently gives a physical way through which the skills may be developed by proposing the associated courses in case of skills deficiency. This goes beyond the recommendation to a definite career pathway towards skill improvement and career placement.

Consolidated Active Job Portal Linkage:

Gap: Foundational papers put the major emphasis on the matching algorithms. Although Akhtar et al. [3] are referring to web crawling of resumes, the author fails to specify the dynamic integration with live job portals (such as JSearchAPI) to retrieve the up-to-date job listing and apply automatically. This forms the lack of connection between the analytical tool and the real job market.

Innovation: To meet this need, our system will directly and immediately integrate with the JSearch API so that live job listings will be accessed. This keeps the job recommendations up to date and also enables the users to move seamlessly to an application page of the recommended job thus, the process becomes much easier and user-friendly.

Extensive User-Friendly Web Interface:

- Gap: Although in the papers under analysis [1], a Streamlit interface is described, the accents are usually on the algorithms in case of Abhishek et al. but in case of the analyzed papers, the main focus is not on the complete approach to develop the intuitive web application, i.e. to integrate all the functions (size of resume, matching of vacancies, recommendation of courses, direct links to the application) into one coherent user experience.

- Innovation: We are working on a one-stop solution consisting of a full-fledged, user-friendly web application (Flask, with Jinja2 templating the frontend and Flask in the backend). It will also have a very interactive and easy to use interface that will enable people to manage their resumes, seek employment and even be advised on the specialized skills they need that will greatly aid in the ease of use and accessibility of the tools as opposed to these being used individually.

VIII. System Requirements

The system must have powerful hardware and software to develop a complete project which would combine AI-powered resume screening, employment suggestions and skill development.

Hardware Requirements:

- A computer containing an Intel Core i5 or AMD Ryzen 5 CPU, and above, to develop and do initial testing. 8 GB RAM to process language models and data (large) effectively.

- It is suggested that NVIDIA GTX 1060 which has 4 GB of VRAM can be used to run BERT models to infer and train much faster and decrease the processing time significantly. The storage should comprise 256GB SSD to store model files, datasets and logs, and optional archives HDD storage.

Software Requirements:

- OS such as Ubuntu Linux 20.04+ should be used in development and deployment because it has a powerful support of AI/ML tool.
- Python as the main programming language.
- Frontend templates using Jinja2 and a dynamic and responsive user interface is flasked.
- Flask for the backend, handling API requests, data processing, and model inference.
- As the backend, the server will receive API requests, process data, and infer endpoints.
- Dependency isolation and management should be done in virtual environments, eg. venv or conda.
- MySQL to store the user data such as credentials and job interests.

Troubleshooting:

- File upload failures: Make sure that the resume is in supported format (.pdf, .docx) and is not over the size limit.
- No Job Met: Expand job search or Revise Resume in more Democratization skills.
- Irrelevant Courses: Please give more information about resume or narrow down on skills.
- Platform Lag: Restart the browser; look into internet/ internet connection. In case of large operations, the processing may be time-consuming.
- Application Issues: Support contact in case of constant problems.

IX. Future Work

In case of further developments, the suggested AI Resume Screener and Job Recommendation system suggests a number of aspects that need improvements that would be necessary to make the system more applicable and closer to the real world.

To improve it, we would recommend more sophisticated capabilities of the system to detect and mitigate against advanced bias. Although our present perspective is on the correct matching, in the future, more advanced methods should be involved to detect and minimize algorithmic bias that may be created due to training data to be able to make more reasonable and fairer recommendations to all users. This is in line with the wider question of AI ethics.

Secondly, one of the critical directions would be to develop finer point skill gap analysis and individual learning journeys. It may mean connecting with the APIs of online learning platforms, and promoting the specific modules/courses dynamically, based on the skill gaps identified by a user and their favorite learning styles. This would further customize and make the course recommendations more effective.

In addition, we will improve the proactive trend analysis of the job market of the system. This includes the crawling and analysis of job market data on a continuous basis to recognize new skills needed in the job market, the jobs that are on the decline, and the changing demands in the industry. The system can give more strategic career advice and skill development recommendations by giving the user a view into these trends before it happens.

The other area of considerable future work is the increase in integrations of job portals and globalization. Although the first target is JSearchAPI, which will provide more specificity to the system, the inclusion of a wider range of international and local job boards would considerably enhance the system. This also involves the investigation of multilingual resumes and job descriptions other than English as well as providing support to a wide range of users.

Lastly, we suggest the creation of a mobile application which is a parallel to the web platform. This would make it easier to access and the system more convenient and accessible through allowing the user to manage his profile, get notifications and look at opportunities wherever they go. This is in line with the growing use of mobile-first solutions with regard to recruitment technology.

X. Workflow

Step 1: User Input: Upload Resumes and Job Preferences.

- The users enter the web-application and post their resumes (e.g., PDF, DOCX).
- The user has the option to key in job preferences (e.g. preferred job role, industry, location).

Step 2: Question Resuming Parsing and Information Extraction.

- The system applies NLP technology to analyze the uploaded resume.
- The skills, experience, education, and contact information are extracted as key information.
- This organized information is then applied to form a detailed profile of the candidate.

Step 3: Retrieving and Generation of Job description.

- Referring to the extracted resume information and user preferences, the system retrieves the corresponding job descriptions on the basis of the integrated job portals (JSearchAPI).
- The BERT embeddings are obtained to represent the semantic meaning of both the text in the resume of the candidate and the job description text.

Step 4: Job- Resume Matching(cosine similarity)

- Each normal resume of the applicant is computed by cosine against the BERT embedding of the job sample description and the BERT embeddings of every job description that is fetched.
- The jobs are ranked depending on the similarity score, which is a percentage state.

Step 5: Course Recommendation & Skill Gap Analysis.

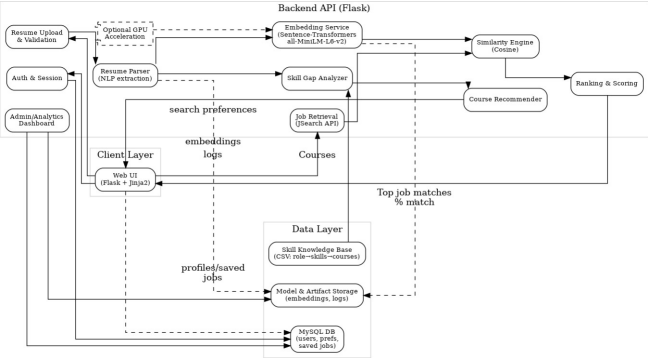
- The system matches the skills applied in the resume of the user with the skills needed in highly matching job responsibilities.
- This system will compare the skills of the user that are detected with a list of skills that are defined in the job category. According to the resulting gaps in skills, the corresponding online course is sought in the same base of knowledge.

Step 6: Output & User Interface

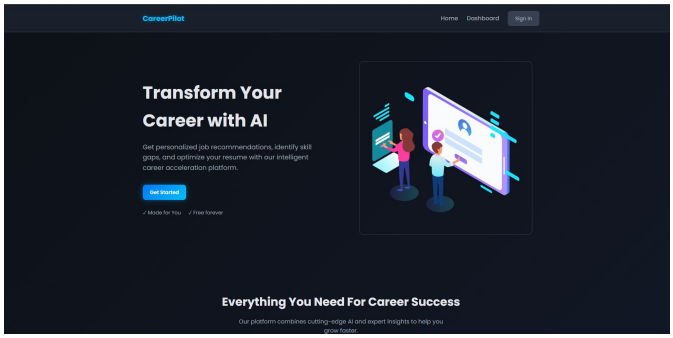
- The user is presented with a customized dashboard by the web application.
- This dashboard includes:
- The list of top job matches with percentage match.
- Direct contact to post job applications on the respective portals.

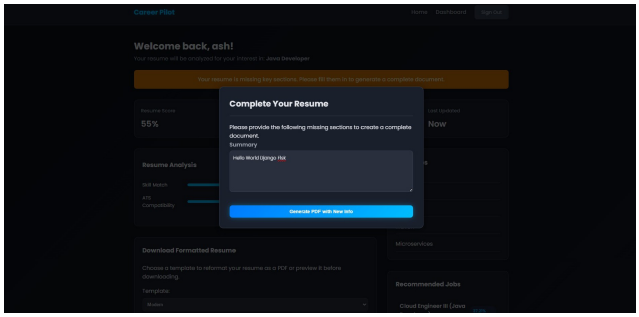
- Recommended courses to take to enhance their profile.
- Users will be able to communicate with the recommendations, save jobs, or learn more about the courses

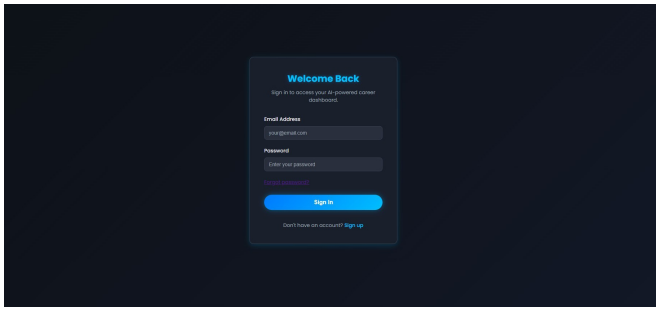
XI. Architecture Diagram

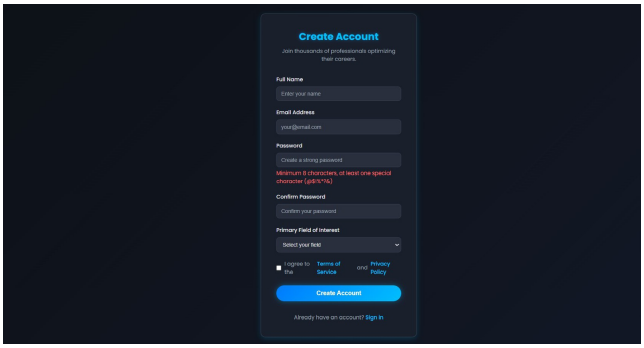


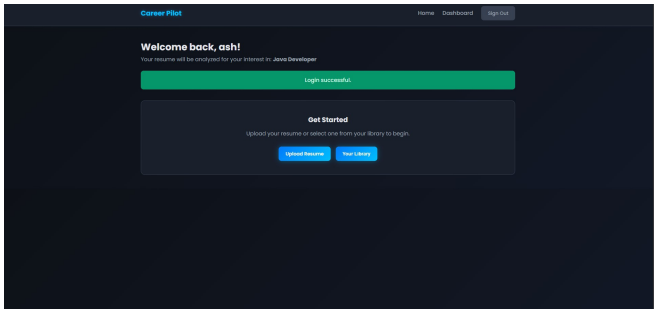
XII. Results and Discussion

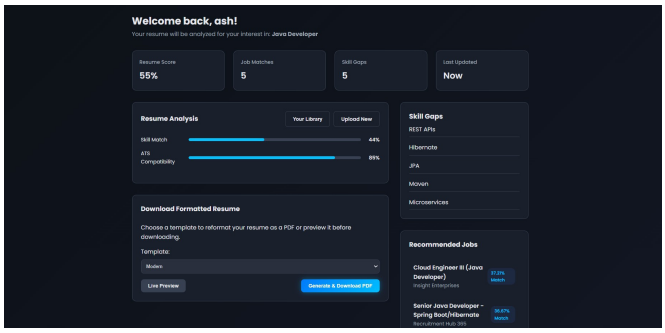


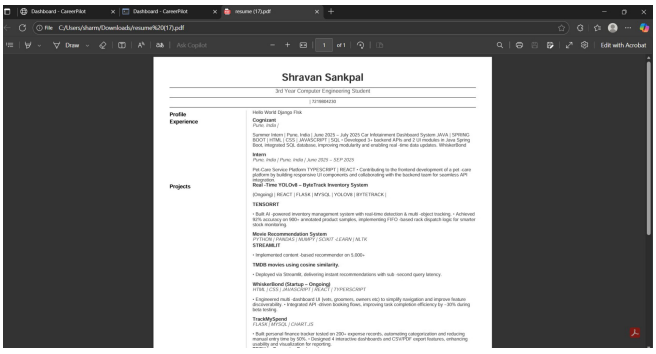












XIII. Conclusion

The suggested system of AI Resume Screener and Job Recommendation could be the next step that will be offered to the modern generation as the next step to reach new heights due to its comprehensive and user-focused approach to career development. Although current competitors usually specialize in small functions, such as bare-bone resume parsing or limited job matching using Cosine similarity, our system is the only one to be a blend of advanced semantic insight with BERT, et al., accurate job matching with Cosine similarity, and practical recommendations as to what courses a user should take to develop the skills required to perform the job. This broad integration together with real-time job portal connection deals with the major limitations of solutions available in the market, which in most cases do not consider the deep case and contextual dimension as well as individual growth trajectories and dynamism.

The unique feature of our project is that it is end-to-end empowering both job seekers and recruiters. To applicants, it turns job search into a guided career path as it does not only provide

opportunities but it also provides the facilities through which one can develop the skills required. To recruiters, it will offer an efficient and precise talent process by matching the most appropriate applicants using smart matching. Our system will be designed to create a new standard of intelligent, integrated and impactful recruitment technology by using the state-of-the-art AI models and focusing our attention on the smooth experience of the user, created through the help of a dedicated web platform. It also goes beyond being just a screening instrument, and becomes a strategic ally in the intricacies of the new job market.

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